

# **Crime in an Analytical World: An Overview of Categorical Data Analysis in Criminology**

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Criminology is an interdisciplinary branch of social science that focuses on various aspects of crime and the legal system. Important areas of criminology include criminal profiling, sentencing decisions, crime prevention, and law creation (Wilson, 2015). One key component that is fundamental to the criminology field is the use of analytical methods. From the very beginning of its conception, criminology has relied on the application of analytical techniques to crime-related data to successfully organize and convey key ideas and information (Tarling, 1986). Criminologists often make use of categorical data in particular. In fact, the use of categorical data is evident in almost every step of the criminological approach.

Exploring the incorporation of categorical data in criminology can help provide insight into what the discipline is about. The aim of this paper is to develop a critical overview of how categorical data have been included in the history, techniques, and applications of criminology, ultimately shaping the discipline as a whole.

### A History of Crime Analysis

The early beginnings of criminology as a discipline can be traced back to the late 18th and early 19th centuries, with controversial figure Cesare Lombroso often credited as one of the first individuals to study criminology (Ferracuti, 1996; Mazzarello, 2011). Lombroso, an Italian physician and eugenicist, was most interested in the causes of criminality. Specifically, Lombroso thought that criminal behaviour was innately predisposed and could be largely identified through an individual's biological and physical features. While Lombroso used analytical methods in his work, these methods were flawed, and he was often criticized for lacking the methodological rigour required for research (Ferracuti, 1996; Mazzarello, 2011; Wolfgang, 1961). Instead, the earliest uses of refined analytical methods in criminology can be largely credited to two figures - Adolphe Quetelet and André-Michel Guerry.

### Adolphe Quetelet (1796-1874)

Adolphe Quetelet, a Belgian mathematician and astronomer, was among the first to apply analytic techniques to crime data. Specifically, Quetelet used statistics and categorical information to analyze patterns in crime data. (Beirne, 1987; Cullen & Wilcox, 2010; Kivivuori, 2024). One of Quetelet's most well-known applications of statistical techniques was his use of correlational analyses on offender data to determine individual factors associated with criminal

behaviours. Through his analyses, Quetelet found that young men who were unemployed, living in poverty, and uneducated, had a higher likelihood of

**Figure 1**  
*Contingency Table presenting Crime Data*

| YEAR         | ACCUSED<br>(TRIED) | CONVICTED | INHABITANTS<br>FOR ONE<br>ACCUSED | CONVICTED<br>FROM 100<br>ACCUSED | ACCUSED OF CRIMES  |                     | RELATIONSHIP<br>BETWEEN THE<br>N OF TYPES<br>OF ACCUSED |
|--------------|--------------------|-----------|-----------------------------------|----------------------------------|--------------------|---------------------|---------------------------------------------------------|
|              |                    |           |                                   |                                  | Against<br>Persons | Against<br>Property |                                                         |
| 1826 .....   | 6,988              | 4,348     | 4,557                             | 62                               | 1,907              | 5,081               | 2.7                                                     |
| 1827 .....   | 6,929              | 4,236     | 4,593                             | 61                               | 1,911              | 5,018               | 2.6                                                     |
| 1828 .....   | 7,396              | 4,551     | 4,307                             | 61                               | 1,844              | 5,552               | 3.0                                                     |
| 1829 .....   | 7,373              | 4,475     | 4,321                             | 61                               | 1,791              | 5,582               | 3.1                                                     |
| Totals ..... | 28,686             | 17,610    | 4,463                             | 61                               | 7,453              | 21,233              | 2.8                                                     |

SOURCE.—Quetelet ([1831a] 1984, p. 20), abridged.

**Note.** This contingency table, created by Adolphe Quetelet (say year), presents statistical data on crime accusations and convictions from the years 1826-1829. Categorical variables include the type of crime ("Against Persons" and "Against Property", and "Year". Sourced from Beirne (1987).

committing crimes and receiving convictions. Taking this a step further, Quetelet broadened his scope of crime analyses to assess environmental factors, leading him to conclude that wealth inequality was highly influential in determining not only the rates of crime, but also what types of crimes were committed (Beirne, 1987; Cullen & Wilcox, 2010). To present his findings, Quetelet commonly made use of contingency tables, which allowed him to neatly organize categorical information (Figure 1) (Beirne, 1987).

### André-Michel Guerry (1802-1866)

Another important figure essential to the development of analytic methods in crime research is André-Michel Guerry, a French lawyer. Like Quetelet, Guerry analyzed crime data to determine the patterns that existed, and how these patterns could inform criminality (Friendly, 2022; Kivivuori, 2024). One of Guerry's earliest published analytical works on crime data was a collaboration with Italian geographer Adriano Balbi. Guerry and Balbi created a cartographic poster, also known as a choropleth map, in which crime rates and education levels were aggregated across geographical areas (Figure 2). This poster successfully allowed direct comparisons of crime data in a visually impactful and innovative way (Friendly, 2022; Kivivuori, 2024).

**Figure 2**

*Choropleth map showing education levels and crime rates per area*



**Note.** This choropleth map, created by André-Michel Guerry and Adriano Balbi (1829), showcases the rates of crime and the level of education across areas in France. Sourced from Kivivuori (2024).

**Figure 3**

*Guerry's table of crime statistics*

| Year              | 1826                    | 1827 | 1828 | 1829 | 1830 | Avg |
|-------------------|-------------------------|------|------|------|------|-----|
| Sex               | All accused (%)         |      |      |      |      |     |
| Male              | 79                      | 79   | 78   | 77   | 78   | 78  |
| Female            | 21                      | 21   | 22   | 23   | 22   | 22  |
| Age               | Accused of theft (%)    |      |      |      |      |     |
| 16-25             | 37                      | 35   | 38   | 37   | 37   | 37  |
| 25-35             | 31                      | 32   | 30   | 31   | 32   | 31  |
| Crime             | Committed in summer (%) |      |      |      |      |     |
| Indecent assault  | .                       | 36   | 36   | 35   | 38   | 36  |
| Assault & battery | .                       | 28   | 27   | 27   | 27   | 28  |

**Note.** This table, created by André-Michel Guerry, shows crime rates segmented by various categorical variables. Sourced from Friendly (2022).

Following this, Guerry continued to expand on his work, using refined methods to analyze new crime-related questions, organize information, and inform criminal policy. Similar to Quetelet, Guerry presented tabulated frequencies of crime, segmented by numerous categorical variables (Figure 3) (Friendly, 2022). These tables demonstrate the early stages of organizing and breaking down crime data in an

easy-to-understand and refined way. Thus, the methods used by both Quetelet and Guerry paved the way for subsequent categorical analytic developments in criminology, ultimately shaping the discipline into what it is today.

### Criminology Methods of Today

The field of criminology has come a long way from what it was during the time of Guerry and Quetelet. Modern-day criminologists have continued advancing the field by developing new and efficient ways to collect, organize, and communicate categorical crime data. These methods range from simple to complex, and are often essential when conducting crime research. Examples of categorical data methods used in criminology include subsetting, mapping, and data scraping.

### Data scraping

During the 19th century, early criminologists and statisticians relied on physical record-keeping to aggregate crime data for their analyses (Shoemaker & Ward, 2017). Now, with the rise of the internet, criminologists rely heavily on information located in online databases (Brewer et al., 2021; Chan & Moses, 2016; Su, 2023). These online databases often contain large volumes of publicly accessible data that require specialized methods to extract relevant information. Specifically, criminologists use a method called “data scraping” to obtain information from online resources (Brewer et al., 2021; Kaplan, 2023; Su, 2023). Typically, criminologists will use automated programs that can quickly and efficiently extract large amounts of data across numerous online sources (Brewer et al., 2021). However, extracting a smaller amount of data, such as data from a single webpage, website, or file, can be done through manual coding methods (Kaplan, 2023). For instance, Kaplan (2023) used features located in the RStudio program, such as the “pdftools” package, to extract table-form border patrol data from a PDF file (Figure 4). For these reasons, the convenience that data scraping offers makes this method an essential first step in many criminological analyses.

**Figure 4**  
Data scraping border patrol data from a PDF

| SECTOR                                | Female        | Male           | Total Apprehensions |
|---------------------------------------|---------------|----------------|---------------------|
| Miami                                 | 219           | 2,061          | 2,280               |
| New Orleans                           | 92            | 828            | 920                 |
| Ranney                                | 65            | 323            | 388                 |
| <b>Coastal Border Sectors Total</b>   | <b>376</b>    | <b>3,212</b>   | <b>3,588</b>        |
| Utah                                  | 97            | 191            | 288                 |
| Buffalo                               | 69            | 378            | 447                 |
| Detroit                               | 78            | 992            | 1,070               |
| Grand Forks                           | 56            | 440            | 496                 |
| Harris                                | 13            | 26             | 39                  |
| Houston                               | 17            | 13             | 30                  |
| Spokane                               | 17            | 191            | 208                 |
| San Antonio                           | 106           | 343            | 449                 |
| <b>Northern Border Sectors Total</b>  | <b>453</b>    | <b>2,574</b>   | <b>3,027</b>        |
| <b>Big Bend/Juarez Sector</b>         | <b>985</b>    | <b>5,917</b>   | <b>6,902</b>        |
| Del Rio                               | 2,622         | 10,894         | 13,516              |
| El Centro                             | 2,791         | 15,842         | 18,633              |
| El Paso                               | 7,364         | 17,829         | 25,193              |
| Laredo                                | 3,651         | 21,809         | 25,460              |
| Rio Grande Valley/Juarez Sector       | 50,306        | 87,256         | 137,562             |
| San Diego                             | 4,117         | 21,969         | 26,086              |
| Tucson                                | 4,693         | 33,964         | 38,657              |
| Yuma                                  | 4,328         | 8,519          | 12,847              |
| <b>Southwest Border Sectors Total</b> | <b>80,057</b> | <b>223,059</b> | <b>303,116</b>      |
| <b>Nationwide Total</b>               | <b>81,086</b> | <b>228,845</b> | <b>310,531</b>      |

  

```

head(final_data)
#           sector agent_staffing
# 1           Miami           111
# 2       New Orleans           63
# 3           Ranney           38
# 4 Coastal Border Sectors Total 212
# 5
# female male
# 1      219 2,061
# 2       92  828
# 3       65  323
# 4      376 3,212
# 5    <NA> <NA>
# 6       97  191

```

**Note.** Left: Segment of the Border patrol data table prior to data scraping. This table was contained in a PDF file. Categorical variables include the sector type and gender. Right: Segments of data that has been scraped from the border patrol table and exported into RStudio. All figures sourced from Kaplan (2023).

## Subsetting

Subsetting is an incredibly useful tool when analyzing criminological data. Subsetting is the practice of reducing a large dataset into one that is smaller and easier to manage (Kaplan, 2023). Due to the nature of the discipline, criminologists typically have to work with large amounts of complex data, often spanning across several years and/or geographical regions, such as crime rates, legal data, and criminal databases (Su, 2023). Reducing the size of a large dataset is essential when focusing on specific questions and analyses. Subsetting funnels out irrelevant information while keeping only information that is necessary for the inquiry at hand (Kaplan, 2023).

**Figure 5**  
*Subsetting code in RStudio*

```
colorado <- filter(ucr, state == "colorado", year %in% 2011:2017)
```

**Note.** This line of code uses the “filter” function within the “dplyr” package to retrieve data from Colorado during the years 2011-2017. Sourced from Kaplan (2023).

Subsetting can easily be accomplished in the RStudio program through various tools, such as using conditional statements. For instance, the “filter” function within the “dplyr” package filters out data based on the conditional statements provided - only information matching the conditional requirements given is included in the output. For example, running the code shown in Figure 5 extracts data from Colorado between the years 2011 and 2017 (Kaplan, 2023).

## Mapping

Another common method of organizing crime-related data is through mapping. Mapping is the process of visualizing information and patterns geographically (Boba, 2001; Bowers et al., 2004; Kaplan, 2023). Mapping is used for several important purposes, such as keeping track of areas with a high amount of criminal activity, and predicting what types of crimes are likely to occur in specific areas (i.e., crime forecasting) (Boba, 2001; Hunt, 2019; Kaplan, 2023). Criminologists have since expanded on the early choropleth mapping works of André-Michel Guerry by creating alternative mapping methods to suit various needs. For instance, a popular modern-day mapping technique is the interactive map (Kaplan, 2023). Traditional methods of mapping tend to limit the information that is presented to keep

**Figure 6**  
*Interactive map of San Francisco marijuana dispensaries*



**Note.** This figure shows a static image of an interactive map visualizing marijuana dispensaries in San Francisco. Some interactive elements include: (a) zoom controls in the top left corner, (b) blue icons to mark the location of each dispensary, and (c) pop-up boxes containing business information for each dispensary. Sourced from Kaplan (2023).

the map clear and easy to understand. Alternatively, interactive maps are able to aggregate a larger amount of data due to their interactive elements (e.g., zooming in and out, selecting different areas, and filtering data), which allows for both broad and specific inquiries (Kaplan, 2023; Ratcliffe, 2002). For example, Figure 6 shows an interactive map created in RStudio using the “leaflet” package. This map includes several interactive features, such as zoom controls, icons to mark relevant locations, and pop-up boxes containing information about these locations that appear when each icon is clicked on. Mapping is thus a powerful tool for visualizing categorical crime data.

### Applications of Categorical Data in Criminology

While categorical data analysis methods provide a deeper understanding of the criminological approach, it is equally important to consider how these methods are applied in various contexts. Due to the interdisciplinary nature of criminology, data analyses are used to assist in many different domains, such as psychology, policy, and law (Wilson, 2015). The wide range of analytical methods available allows both criminologists and non-criminologists to benefit from these techniques and use them to address specific topics and questions. Examples of criminological categorical data applications include offender typologies, sentencing decisions, risk assessments, refugee claims, and cybercrimes.

### Offender Typology

Assessing the determinants of criminal behaviour has been a prevalent goal throughout the history of criminology. As previously mentioned, early criminologists and statisticians used data analysis to determine what individual and environmental characteristics were common across offenders (Ferracuti, 1996; Friendly, 2022; Kivivuori, 2024; Mazzarello, 2011). These analyses help create what is known today as “offender profiles/typologies”, which connect specific criminal behaviours to various characteristics (Byrne & Roberts, 2007). Offender typologies can help inform other important areas of criminology, such as risk assessments and sentencing decisions.

To create offender typologies, criminologists aggregate categorical data into meaningful

variable groups, then assess the relationships and patterns among these variables (Byrne & Roberts, 2007). A simple way to do this is through organizing data

**Figure 7**  
Offender characteristics across latent classes

ANOVA comparison of offender and prior offense characteristics across classes.

|                                           | Coercive child molester<br>M (SD) | Sadistic rapist<br>M (SD) | Stranger focused<br>M (SD) | F-value                 | $\eta^2$ |
|-------------------------------------------|-----------------------------------|---------------------------|----------------------------|-------------------------|----------|
| Child molestations                        | 4.35 (3.52)                       | 1.27 (2.27)               | 1.71 (4.36)                | 7.73 <sup>***,a</sup>   | 0.150    |
| Coercive child molestations               | 3.10 (3.32)                       | 0.55 (1.30)               | 0.86 (3.09)                | 7.85 <sup>***,ab</sup>  | 0.150    |
| Forcible child molestations               | 1.22 (2.62)                       | 0.73 (1.24)               | 0.86 (3.14)                | 0.36                    | 0.008    |
| Rapes <sup>a</sup>                        | 0.18 (0.57)                       | 1.50 (1.44)               | 1.29 (1.31)                | 16.30 <sup>***,ab</sup> | 0.268    |
| Non-physical sexual offenses <sup>a</sup> | 0.29 (0.96)                       | 0.14 (0.47)               | 1.19 (2.06)                | 4.99 <sup>**</sup>      | 0.101    |
| PCL-R total                               | 27.02 (6.69)                      | 29.73 (3.80)              | 30.71 (6.87)               | 1.33                    | 0.067    |
| PCL-R Factor 1 percentile                 | 67.44 (7.01)                      | 69.03 (8.87)              | 74.90 (12.54)              | 0.14                    | 0.010    |
| PCL-R Factor 2 percentile                 | 55.52 (6.71)                      | 60.90 (8.66)              | 67.00 (11.62)              | 2.69                    | 0.172    |

**Note.** This ANOVA table displays offender characteristics across classes of sexual offenders, as revealed from latent class analyses. Sourced from Kaseweter et al. (2016).

in contingency tables, as demonstrated in the early works of both Guerry and Quetelet (Friendly, 2022; Kivivuori, 2024). However, criminologists have since begun utilizing more complex techniques to explore different aspects of criminal populations. For instance, latent class analysis has become an increasingly popular way of determining criminal typology (Belyea et al., 2025; Kaseweter et al., 2016). Latent class analysis is an analytical method that identifies clusters of previously unrevealed (latent) groups based on a set of specific factors (Belyea et al., 2025;

Weller et al., 2020). Latent class analysis is especially useful for discovering the behaviours and characteristics of very specific offender subgroups. For instance, Kaseweter et al. (2016) used latent class analysis to explore specific variables related to sexual offending, such as weapon use, childhood abuse, and sexual fantasy types. The results of these analyses revealed typologies for several previously unexplored subgroups of sexual offenders (Figure 7). Another study by Belyea et al. (2025) conducted latent class analyses with levels of violent behaviour

indicators (e.g., substance use, mental health, social influences, etc) (Figure 8) to determine (a) the typologies of violent female perpetrators, and (b) the likelihood of reoffending across each typology. These studies demonstrate the importance of using enhanced categorical analysis methods to determine various offender characteristics.

**Figure 8**  
Variables related to violent offending in women

| SPin Dynamic Need Domain                 | Overall Rating |      |       |      |          |      |       |      |
|------------------------------------------|----------------|------|-------|------|----------|------|-------|------|
|                                          | None           |      | Low   |      | Moderate |      | High  |      |
|                                          | n              | %    | n     | %    | n        | %    | n     | %    |
| Substance use                            | 1,250          | 33.1 | 703   | 18.6 | 1,557    | 41.3 | 263   | 7.0  |
| Social influences                        | 1,384          | 36.7 | 751   | 19.9 | 1,078    | 28.6 | 560   | 14.8 |
| Family                                   | 561            | 14.9 | 803   | 21.3 | 1,412    | 37.4 | 997   | 26.4 |
| Employment                               | 2,400          | 63.6 | 636   | 16.9 | 500      | 13.3 | 237   | 6.3  |
| Attitudes                                | 2,309          | 61.2 | 958   | 25.4 | 355      | 9.4  | 151   | 4.0  |
| Social/Cognitive skills                  | 2,731          | 72.4 | 594   | 15.7 | 239      | 6.3  | 209   | 5.5  |
| Stability                                | 1,556          | 41.2 | 1,004 | 26.6 | 917      | 24.3 | 296   | 7.8  |
| Mental health (static and dynamic flags) | N/A            | N/A  | 1,446 | 38.3 | 644      | 17.1 | 1,686 | 44.6 |

*Note.* This table displays levels of numerous variables related to violent offending in women. These variables were used in latent class analyses to determine typologies of violent female offenders. Sourced from Belyea et al. (2025).

## Sentencing Decisions and Risk Assessments

The importance of categorical analyses extends beyond theoretical research questions and into real-world decision-making. One prominent real-world application is the use of analyses to determine offender sentencing (Martínez-Garay, 2023). Sentencing decisions often rely on several important factors, such as criminal history, recidivism (reoffending) likelihood, and crime type (Martínez-Garay, 2023; Sheeran, 2025). There are many methods for exploring sentencing decisions. As previously mentioned, offender typologies are often used to help assess the odds of reoffending based on specific characteristics and behavioural patterns (Belyea et al., 2025). Additionally, actuarial risk assessments aggregate both dynamic (e.g., personality, substance use, social support, etc) and static (e.g., offending history) data to determine an offender's recidivism risk level (Fass et al., 2008; Tim Dixon, 2000).

**Figure 9**

Variables related to sentencing decisions in South Carolina

| Variable                        | Description                                                                                                         | Code                                                                                                                                   | Summary (%)                                   | Statistics (n)                       |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|--------------------------------------|
| <i>Dependent variable</i>       |                                                                                                                     |                                                                                                                                        |                                               |                                      |
| Prison term                     | Expected minimum # of months sentenced (0 to 720 months)                                                            |                                                                                                                                        | Mean = 16.2 Std. Dev. = 64.4                  |                                      |
| <i>Case characteristics</i>     |                                                                                                                     |                                                                                                                                        |                                               |                                      |
| Offense seriousness             | 5-level ordinal score From the S.C. Crime Classification Scheme                                                     | 1 = Misdemeanor<br>2 = Felony (Class F)<br>3 = Felony (Class E)<br>4 = Felony (Class D)<br>5 = Felony (Class A, B, C, or Unclassified) | 15.3 %<br>46.0 %<br>19.6 %<br>11.0 %<br>8.2 % | 2694<br>8136<br>3455<br>1942<br>1444 |
| Commitment score                | 12-level ordinal measure Number of commitment offenses                                                              | 1 = Less serious<br>12 = Very Serious                                                                                                  | Mean = 1.9 Std. Dev. = 1.8                    |                                      |
| Offense type                    | 4-category nominal indicator of the classification of crime committed (Violent offenses are the reference category) | 1 = Violent<br>2 = Drug<br>3 = Property<br>4 = Other                                                                                   | 34.3 %<br>14.5 %<br>33.2 %<br>18.0 %          | 6065<br>2561<br>5858<br>3187         |
| Mandatory minimum               | Minimum prison sentence mandated                                                                                    | 1 = Yes<br>0 = No                                                                                                                      | 5.1 %<br>94.9 %                               | 897<br>16,746                        |
| Trial                           | Found guilty after trial                                                                                            | 1 = Guilty after Trial<br>0 = Guilty Plea                                                                                              | 1.5 %<br>98.5 %                               | 258<br>17,413                        |
| <i>Offender characteristics</i> |                                                                                                                     |                                                                                                                                        |                                               |                                      |
| Criminal history                | 5-level ordinal score Derived from the number and severity of prior offenses                                        | 1 = None<br>2 = Minimal<br>3 = Moderate<br>4 = Considerable<br>5 = Extensive                                                           | 36.6 %<br>32.9 %<br>17.4 %<br>5.9 %<br>7.2 %  | 6460<br>5806<br>3080<br>1048<br>1277 |
| Black                           | Race                                                                                                                | 1 = Black<br>0 = White                                                                                                                 | 62.0 %<br>38.0 %                              | 10,990<br>6721                       |
| Male                            | Gender                                                                                                              | 1 = Female<br>0 = Male                                                                                                                 | 16.6 %<br>83.4 %                              | 2929<br>14,742                       |
| Age                             | Age (15 to 81 years old)                                                                                            |                                                                                                                                        | Mean = 31.3 Std. Dev. = 10.1                  |                                      |

**Note.** This table lists numerous variables that were factored into sentencing decisions in South Carolina. Racial biases were revealed when analyzing this data using a hurdle regression model. Sourced from Hester & Hartman (2016).

Categorical data is also an important component in retrospective research on sentencing decisions. Often, criminologists will examine historical data on sentencing decisions to look for evidence of patterns or biases. For instance, a study by Hester & Hartman (2016) analyzed various factors of sentencing decisions in South Carolina (Figure 9) using a hurdle regression model to determine sentencing fairness. Results showed that: (a) Black offenders with low criminal histories received more convictions than their White counterparts, and (b) Black offenders convicted of less-severe crimes received longer sentences than their White counterparts (Hester & Hartman, 2016). Thus, categorical data is useful when considering past, present, and future sentencing and risk assessment decisions.

## Other Applications in Criminology Cybersecurity

Crime is not just a real-world phenomenon, and the internet has now paved the way for new forms of crime to emerge. Specifically, criminals now use the internet as a tool to carry out criminal acts (Hawdon, 2021; Phillips et al., 2022). These criminal acts, termed “cybercrimes”, continue to grow in popularity, thus requiring constant vigilance from criminologists and law enforcement to ensure that these crimes are dealt with. The most popular method of monitoring the internet for cybercrimes is data scraping, which allows criminologists to comb through a vast amount of data sourced from the internet to find illicit material and behaviours (Brewer et al., 2021). Analytical tools are also important in preventing cybercrimes from occurring. For example, offender typologies are often used to determine what factors increase the risk of an individual committing a cybercrime (Bekkers et al., 2025). Although there is controversy surrounding digital monitoring and data scraping due to concerns over heightened surveillance and a lack of privacy, criminologists argue that these methods are necessary to ensure the safety of internet users (Brewer et al., 2021; Phillips et al., 2022).

## Refugee Law

Criminology is not just limited to offender statistics and criminal data - many areas of criminology deal with legal proceedings and policies as well. For instance, refugee claims depend on the examination of various characteristics that increase a claimant’s risk of danger and/or prosecution in their origin country, such as race, gender, and religion (Béchar & Mason, 2020). Although there are guidelines in place for refugee claims and appeals, these decisions are often influenced by the year, the country in which the claim was filed for, and/or the judge

making the decisions. As this information is public, many criminologists and academic researchers have aggregated refugee decision data to look for patterns. An example of this is the Refugee Law Lab Portal, created by Osgoode Hall Law School associate professor Sean Rehaag (2024). The Refugee Law Lab Portal is a website containing data on Canadian refugee decisions dating back to 1993, organized by country of origin, year, and judge (Sean Rehaag, 2024). The interactive elements of the website allow for easy visualization of trends and efficient organization of both categorical and quantitative data.

### **Conclusion**

Criminology is a vast and complex field that focuses on interdisciplinary areas of crime, research, law, and policy (Wilson, 2015). Criminology has been continuously changing and evolving since its early beginnings during the late 18th and 19th centuries to adapt to the progression of crimes and society. However, one key component that has been consistently essential in almost every aspect of the field is the use of categorical data. The current paper highlights the importance of categorical data in criminology by exploring the history of criminological data analyses, the analysis methods of today, and various applications of categorical data throughout the field. Exploring these methods and applications of categorical data allows us to better understand the development, goals, and motivations of the field of criminology.

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